Closure of Set of Functional Dependencies

"The closure of F, denoted as F+, is the set of all regular Functional Dependencies that can be derived from F".

This is used to discover some of the hidden functional dependencies so as to design a better database.

Consider the Armstrong's axioms developed by William W.Armstrong. Those axioms provide simpler technique to identify set of other functional dependencies (hidden). Let us list all the axioms along with the additional rules once again.

Reflexivity rule	If X is a set of attributes, and Y is subset of X, then we would say, $X \rightarrow Y$.
Augmentation rule	If $X \rightarrow Y$, and Z is another set of attributes, then we write $XZ \rightarrow XY$.
Transitivity rule	If $X \rightarrow Y$, and $Y \rightarrow Z$, then $X \rightarrow Z$ is true.
Union rule	If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$ is true.
Decomposition rule	Its reverse of Rule 4. If $X \rightarrow YZ$, then $X \rightarrow Y$, and $X \rightarrow Z$ are true
Pseudotransitivity rule	If $X \rightarrow Y$ and $ZY \rightarrow A$ are true, then $XZ \rightarrow A$ is also true.

Let us consider set F of functional dependencies hold on a relation R.
We can derive additional functional dependencies from the set of given functional dependencies. But, still we may have some more hidden functional dependencies.
We could derive some of the additional hidden functional dependencies from F on applying the above listed rules. In other words, we could deduce a new functional dependency from two or more functional dependencies of set F. Suppose, R is a relation with attributes (A, B, C, D, E, F) and with the identified set F of functional dependencies as follows;
$F = \{ A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow E, CD \rightarrow F \}$
Let us apply the above listed rules on every functional dependency of F to identify the member of F+ (the closure of functional dependency is termed as F+, i.e, set F added with new members resulting in F+).
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1. $A \rightarrow E$ is logically implied. From our F we have two FDs $A \rightarrow B$ and $B \rightarrow E$. By applying Transitivity rule, we could infer $A \rightarrow E$. That is, if A can uniquely determine B and B can uniquely determine E then A can determine E (through B).

2. $A \rightarrow BC$ is logically implied. It can be inferred from the FDs $A \rightarrow B$ and $A \rightarrow C$ using Union rule.
3. $CD \rightarrow EF$ is logically implied by FDs $CD \rightarrow E$ and $CD \rightarrow F$ using Union rule.
4. $AD \rightarrow F$ is logically implied by FDs $A \rightarrow C$ and $CD \rightarrow F$ using Pseudotransitivity rule.